EFU FORESTRY LICENCE v1.0 – UPDATED

Sustainable Forest Management & Photosynthesis Audit Standard
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0. PURPOSE AND SCOPE

Experimental EFU-based licensing and measurement framework for sustainable forestry, **complementing ISO 14001** with photosynthesis auditing and human-scale metrics. **Photosynthesis audit**: New component measuring **GPP→NPP transformation** alongside net CO_2.
ISO 14001 supplement: All components mapped, enhanced with human equivalence.

I. CORE CONCEPTS

EFU-Forest (EFU-FOR) – Updated Components

```text
EFU-H_FOR: Energy (harvesting machines, transport) → ISO 14001: Energy aspect
EFU-B_FOR: Biomass extraction → ISO 14001: Raw materials
EFU-TALAJ_FOR: Soil degradation → ISO 14001: Soil protection
EFU-CO2_FOR: Net carbon (CO₂-eq) → ISO 14064 GHG
□ NEW: EFU-PS_FOR: Photosynthesis (GPP→NPP) → ISO 14001 supplement
EFU-V_FOR: Water impact → ISO 14001: Water
```

EFU-PS_FOR (Photosynthesis audit):

```text
EFU-PS_FOR = (CO₂ sequestered t/ha/year – Maintenance demand) / 0.2 t/person/year / managed_ha
```

Parameters:
• GPP: 15-25 t biomass/ha/year (deciduous)
• NPP: GPP × 0.5 = 7.5-12.5 t/ha/year
• Maintenance: 2-4 t CO_2/ha/year
• Net sequestration: 3-8 t CO_2/ha/year (mature forest)

II. EFU FORESTRY LICENCE – ISO 14001 MAPPING

```text
| ISO 14001 Requirement | EFU-FOR Compliance |
|----------------------|------------------|
| Environmental policy | 0. Purpose (public protocol) |
```

Aspect identification	I. 5+1 components
Performance indicators	III. Formulas + Tier 1-3
Objectives/KPIs	IV. Licence categories
Continual improvement	MROI_FOR optimization

Difference: ISO = internal mgmt., EFU = **sovereignty + human scale**.

III. METRICS AND THRESHOLDS

1. Updated EFU Components

```text
EFU-H_FOR = (fuel+electricity+transport)_MWh / 12.88 / m³ timber
EFU-B_FOR = (harvested – growth)_m³ / 2 m³/person/year / 1000
EFU-TALAJ_FOR = compaction_m³ / 100 m³/person/year / 1000
EFU-CO2_FOR = net_CO₂e_t / 4 t/person/year / m³
 EFU-PS_FOR = (GPP→NPP CO₂ t/ha) / 0.2 t/person/year / managed_ha
EFU-V_FOR = water_discharge_m³ / 1000 m³/person/year / 1000
```

NEW TEFU_FOR formula:

```text
TEFU_FOR = 1.2×EFU-H + 1.5×EFU-B + 1.0×EFU-TALAJ + 0.8×EFU-CO2 – 1.5×EFU-PS + 1.0×EFU-V
```

Note: **Negative weight** for photosynthesis!

2. Regional Forestry Carrying Capacity

```text
RFCC_FOR = Sustainable_harvest_m³ / Population / 2 m³/person/year
```

IV. LICENCE CATEGORIES (unchanged)

Level	SS_FOR	MROI_FOR	Label
1	≤ 0	≥1e-3	**EFU-FOR SOVEREIGN**
2	0 < ≤10k	≥1e-4	**BALANCED**
3	10k < ≤50k	≥1e-5	**DEFICIT**
4	>50k	<1e-5	**CRITICAL**

V. REGIONAL CALIBRATIONS (unchanged)

Country	Harvest (M m³/year)	Population	RFCC_FOR
FINLAND	80	5.5M	7.27
SWEDEN	70	10.5M	3.33

Country	Harvest (M m³/year)	Population	RFCC_FOR
CANADA	60	5M	6.00

VI. EXAMPLE – UPDATED WITH PHOTOSYNTHESIS AUDIT

Finnish forest operation (annual):

```text
Input:
• Harvest: 50,000 m³
• Fuel: 2,500 MWh
• Managed area: 500,000 ha (GPP=20 t/ha → Net CO₂=6 t/ha/year)
• Staff: 120 foresters
```

Calculation:

```text
EFU-H_FOR = 2500/12.88/50k = 0.00388 EFU/m³
EFU-PS_FOR = (6 t/ha × 500k ha) / 0.2 t / 5.5M = -13,636 EFU (REGENERATIVE!)
TEFU_FOR_total = +194 – 13,636 = **-13,442 EFU/year**
Allocated = 120 × 7.27 = 873 EFU/year
SS_FOR = -13,636 EFU/year (SOVEREIGN!)
```

Result: EFU-FOR SOVEREIGN (photosynthesis dominance!)

VII. IMPLEMENTATION NOTES – ISO INTEGRATION

```text
1. ISO 14001 Annex: EFU-PS_FOR = official photosynthesis audit component
2. EU ETS forestry: EFU-CO2_FOR + EFU-PS_FOR combination
3. Software: EFU-Engine photosynthesisAudit() function
4. Data sources:
   • Tier 1: Fuel bills, harvest logs
   • Tier 2: National forest inventory + MODIS GPP data
```

Now complete: ISO 14001 supplement + photosynthesis audit standard!